Constructor University Bremen

MPFR LAB II

Part 2: A Brushed DC Motor Model
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Task1.4 - Task 1.9

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Table 1.1: Parameters for the Pololu 19:1 Metal Gearmotor at 12V. The values are at the output-shaft which includes the effect of the gear-box.

Parameter	**Value**	**In SI Units**	**Description**
Free-run speed	500 RPM	52.36 rad/s	ω_0, the shaft-speed at no-load
Free-run current	300 mA	0.3 A	i_0, the no-load current
Stall current	5000 mA	5 A	i_s
Stall torque	84 oz-in	0.5932 Nm	τ_s

Table 1.2: Parameter-variation across models, as the gear-ratio is varied. All values are at 12V.

Gear-ratio	ω_0	τ_s	i_s
1:1	11000 RPM	5 oz-in	5 A
19:1	500 RPM	84 oz-in	5 A
30:1	350 RPM	110 oz-in	5 A
50:1	200 RPM	170 oz-in	5 A
70:1	150 RPM	200 oz-in	5 A
100:1	100 RPM	220 oz-in	5 A
131:1	80 RPM	250 oz-in	5 A

Task 1.4

Estimate k_t SI units for this motor (Table 1.1) from the data provided using (1.4), the stall current and the stall torque.

$$\begin{aligned} k_t &= \frac{T_g}{i} \\ &= \frac{0.5932}{5} \\ &= 0.11864 \text{ Nm A}^{-1} \end{aligned}$$

Task 1.5

Estimate k_e in Si units for this motor from the data provided using (1.7). We will get a better estimate later.

$$\begin{aligned} k_e &\leq \frac{v_a}{\omega_0} \\ &\leq \frac{12}{52.36} \\ &\leq 0.22918 \text{ V s rad}^{-1} \end{aligned}$$

Task 1.6

Estimate b in SI units for the motor in Table 1.1.

$$\begin{aligned} \mathrm{b} &= \frac{\mathrm{k_t} \times i_0}{\omega_0} \\ &= \frac{0.11864 \times 0.3}{52.36} \\ &= 6.798 \times 10^{-4}\ \mathrm{N\ m\ s\ rad^{-1}} \end{aligned}$$

Task 1.7

Compute R_a from the motor data given. Using this value, obtain a better estimate for k_e than the one we obtained earlier in (1.7)? Hint: consider (1.18) in the no-load steady state condition.

$$\begin{aligned} \mathrm{R_a} &= \frac{\mathrm{v_a}}{\mathrm{i_s}} \\ &= \frac{12}{5} \\ &= 2.4\ \Omega \end{aligned}$$

With a steady state,

$$\mathrm{v_a} - \mathrm{i}R_a - L_a \frac{di}{dt} = k_e \theta_g$$

$$\mathrm{v_a} - \mathrm{i}R_a - 0 = k_e \theta_g$$

$$\begin{aligned} k_e &= \frac{\mathrm{v_a} - \mathrm{i}R_a}{\theta_g} \\ &= \frac{12 - 0.3 \times 2.4}{52.36} \\ &= 0.21543\ \mathrm{V\ s\ rad^{-1}} \end{aligned}$$

Task 1.8

On a sheet of paper, derive (1.21b) from (1.17) and (1.20) as explained. These transfer-functions Hϑv(s) and Hϑ`(s) are important for position-control (a.k.a. servo mode) of the motor.

From (1.17), $(Js^2+bs)\Theta_g(s) = k_t I(s) - T_L(s)$

$$I(s) = \frac{(Js^2+bs)\Theta_g(s) + T_L(s)}{k_t} \quad \text{—— ①}$$

From 1.20, $V_a(s) - (R_a + L_a s) I(s) = k_e s \Theta_g(s)$

$$I(s) = \frac{V_a(s) - k_e s \Theta_g(s)}{(R_a + L_a s)} \quad \text{—— ②}$$

From ① and ②, $\dfrac{(Js^2+bs)\Theta_g(s) + T_L(s)}{k_t} = \dfrac{V_a(s) - k_e s \Theta_g(s)}{R_a + L_a s}$

$$(Js^2+bs)\Theta_g(s)(R_a+L_a s) + T_L(s)(R_a+L_a s) = V_a(s) k_t - k_e s \Theta_g(s) k_t$$

$$\Theta_g(s)(Js^2+bs)(R_a+L_a s) + \Theta_g(s) k_e k_t s = V_a(s) k_t - T_L(s)(R_a+L_a s)$$

$$\Theta_g(s) s \left[(Js+b)(R_a+L_a s) + k_e k_t\right] = V_a(s) k_t - T_L(s)(R_a+L_a s) \quad \text{—— ③}$$

From 1.21a, $D(s) = (R_a + L_a s)(Js+b) + k_e k_t$ —— ④

Sub ④ into ③, $\Theta_g(s) s D(s) = V_a(s) k_t - T_L(s)(R_a + L_a s)$

$$\Theta_g(s) = \frac{V_a(s) k_t - (R_a + L_a s) T_L(s)}{s D(s)}$$

$$= \frac{k_t}{sD(s)} V_a(s) - \frac{(R_a + L_a s)}{sD(s)} T_L(s) \quad \text{—— (1.21b)}$$

Task 1.9

Show that if the inductance La is small and can be ignored, Hwv(s) can be written as a first order system

$$H_{\omega v}(s) \approx \frac{K}{s+\alpha}$$

what are αand K in terms of the other parameters?

$H_{\omega v}(s) = \frac{k_t}{D(s)}$ ---------- 1

$D(s) = (R_a + L_a s)(Js + b) + k_e k_t$ ----------2

From 1 and 2,

$$H_{\omega v}(s) = \frac{k_t}{(R_a + L_a s)(Js + b) + k_e k_t}$$

$$= \frac{k_t}{(R_a + 0)(Js + b) + k_e k_t}$$

$$= \frac{k_t}{R_a Js + R_a b + k_e k_t}$$

$$= \frac{\frac{k_t}{R_a J}}{s + \frac{R_a b}{R_a J} + \frac{k_e k_t}{R_a J}}$$

By comparing,

$$K = \frac{k_t}{R_a J}$$

$$\alpha = \frac{R_a b}{R_a J} + \frac{k_e k_t}{R_a J}$$

$$\alpha = \frac{b}{J} + \frac{k_e k_t}{R_a J}$$